

Ultimate obstination, constructively

a machine-checked proof and its computational content

Abstract

This note accompanies an Agda formalisation of Coquand’s *Une preuve directe du Théorème d’Ultime Obstination* (`min1.pdf`), a direct constructive proof of Colson’s ultimate-obstination theorem. We recall the statement, sketch the proof at a high level, and explain its computational pay-off: the Scott-continuous extension of a primitive-recursive term is a computable object, so one can *evaluate* it, in particular at the infinite diagonal point $x = S^\omega(\perp)$. We illustrate this with a primitive-recursive min, which returns $\perp$ on $(S^\omega(\perp), S^\omega(\perp))$ — obstination in one line. Every statement below is machine-checked with `agda -safe -without-K -exact-split`, with no postulates, holes, or pragmas.

1 The domain and primitive recursion

Let D be the domain of *lazy naturals*: its elements are the complete numbers $S^k(0)$, the incomplete finite ones $S^k(\perp)$ (with $S^0(\perp) = \perp$), and the single infinite element $S^\omega(\perp)$. The order is generated by $S^k(\perp) \leq S^\omega(\perp)$ and $S^k(\perp) \leq S^l(x)$ for $k \leq l$; the elements $S^k(0)$ and $S^\omega(\perp)$ are maximal. Write $F \subseteq D$ for the finite elements.

PR is the usual syntax of primitive-recursive functions: the constant 0, the projections, the successor S , and closure under composition and primitive recursion

$$f(0, Y) = g(Y), \quad f(Sx, Y) = h(x, f(x, Y), Y).$$

Each PR term of arity n denotes a monotone map $F^n \rightarrow F$, written $\text{ev}(p)$; moreover every such map is *stable* (Berry): if A, B have a common upper bound then $f(A \wedge B) = f(A) \wedge f(B)$. Both facts are proved by a direct induction.

2 The property and the theorem

Definition 1 (Ultimate obstination). *A monotone $f: F^n \rightarrow F$ satisfies the property at $A \in D^n$ when there is a finite $A_0 \leq A$ such that one of:*

1. *there is m with $f(X) = S^m(0)$ for all finite $X \geq A_0$;*
2. *there is m and a coordinate i with $A(i)$ incomplete finite, $A_0(i) = A(i)$, and $f(X) = S^m(\perp)$ for all finite X with $X(i) = A_0(i)$ and $X[i] \geq A_0[i]$;*
3. *there is a coordinate i with $A(i) = S^\omega(\perp)$, an integer k with $A_0(i) = S^k(\perp)$, and a numeric φ (constant or strictly increasing from k) with $f(X) = S^{\varphi(m)}(\perp)$ for all finite X with $X(i) = S^m(\perp)$, $k \leq m$, $A_0[i] \leq X[i]$.*

The existential and disjunctive content is read intuitionistically. We write $\text{UO } f A$ for a witness, and say f satisfies ultimate obstination if it does so at every point.

In the formalisation this is a three-constructor datatype over the three cases (file `Property.agda`).

Proposition 1 (Colson; Coquand’s note). *Every element of PR satisfies ultimate obstination.*

In Agda (`Prop1.agda`):

$$\text{prop1} : (p : \text{PR}) (A : \text{Type}) \rightarrow \text{Wf } p (\text{length } A) \rightarrow \text{UO } (\text{ev } p) A,$$

where $\text{Wf } p n$ records that p is well-formed of arity n (e.g. a projection π_i needs $i < n$; $\text{prec } g h$ at arity $n = m+1$ needs g of arity m and h of arity $m+2$).

Proof sketch. Induction on p .

- *Constants, projections, successor* satisfy the property outright (each is eventually constant or reads a single coordinate).
- *Composition.* The witness functions φ are closed under composition, so $g \circ \langle h_1, \dots, h_k \rangle$ is obstinate whenever the parts are.
- *Primitive recursion* is the heart. For a finite first argument one recurs on its height. For the infinite first argument $x = S^\omega(\perp)$ one forms the Kleene approximation

$$u_0 = \perp, \quad u_{k+1} = \hat{h}(S^\omega(\perp), u_k, Y),$$

which is increasing and, once $u_k = u_{k+1}$, constant thereafter. A constructive dichotomy decides, at each stage n , whether the sequence has already stabilised (giving a finite limit, hence case 1 or 2) or still satisfies $S^n(\perp) \leq u_n$. In the second regime one uses that h is obstinate at $(S^\omega(\perp), S^\omega(\perp), Y)$ and *stable* to pin the growth to a single coordinate and read off a witness ψ (with ψ constant or strictly increasing), yielding case 3. Stability — not the stronger Kahn–Plotkin sequentiality — is exactly what excludes two coordinates independently driving f to infinity.

The recursion argument is carried out once over an abstract base/step pair; the induction then feeds it the *arity-guarded* interpretations of the sub-terms (padded to be total and obstinate) and transports the result back to the concrete interpreter. $\square$

3 Computational content: evaluating $\hat{f}$

The point of phrasing obstination intuitionistically is that a witness *carries the value* of the Scott-continuous extension. Here m , i , φ and k are the data recorded by the witness of Definition 1: m is the constant of case 1/2, and in case 3 (the infinite coordinate i) k is the threshold with $A_0(i) = S^k(\perp)$ and φ is the accompanying numeric function, which is *either* constant *or* strictly increasing from k — case 3 does not fix which; the two sub-cases are read off separately. Since $A(i) = S^\omega(\perp) = \sup_{m \geq k} S^m(\perp)$ and $f(X) = S^{\varphi(m)}(\perp)$ on the witnessed region, continuity gives $\hat{f}(A) = \sup_{m \geq k} S^{\varphi(m)}(\perp)$, which is $S^{\varphi(k)}(\perp)$ when φ is constant (then $\varphi(m) = \varphi(k)$ for all $m \geq k$) and $S^\omega(\perp)$ when φ is strictly increasing. Thus

$$\text{val}(\text{UO } f A) = \begin{cases} S^m(0) & \text{case 1,} \\ S^m(\perp) & \text{case 2,} \\ S^{\varphi(k)}(\perp) & \text{case 3, } \varphi \text{ constant (a fixed finite value),} \\ S^\omega(\perp) & \text{case 3, } \varphi \text{ strictly increasing,} \end{cases}$$

which is the value $\hat{f}(A)$ of the extension (file `Property.agda`, `uoValue`).

Corollary 1 (Computability). *For every PR term f of arity n the extension $\hat{f}: D^n \rightarrow D$ is computable: $\hat{f}(A) = \text{val}(\text{prop1 } f \ A)$ is a total, effective function of A .*

Since `prop1` and `val` are total, postulate-free Agda functions, their composition is the algorithm (file `Computable.agda`):

$$\mathbf{fhat} : (f : \text{PR})(A : \text{Tup}) \rightarrow \text{Wf } f \ (\text{length } A) \rightarrow D.$$

In particular one may evaluate at the *infinite diagonal* $x = S^\omega(\perp)$ in every coordinate; call this `fhat-diag` $f \ n$. These are not symbolic answers: for concrete f they reduce, inside the type-checker, to a concrete element of D .

4 pred and min at $S^\omega(\perp)$

Take the standard primitive-recursive terms (file `PredMin.agda`):

$$\begin{aligned} \text{pred} &= \text{prec } 0 \ \pi_0 & \text{pred}(0) &= 0, \ \text{pred}(S \ x) = x, \\ \text{sub} &= \text{prec } \pi_0 \ (\text{pred} \circ \pi_1) & \text{sub}(y, x) &= x \dot{-} y, \\ \text{min} &= \text{sub} \circ \langle \text{sub} \circ \langle \pi_1, \pi_0 \rangle, \ \pi_0 \rangle & \text{min}(x, y) &= x \dot{-} (x \dot{-} y), \\ \text{add} &= \text{prec } \pi_0 \ (S \circ \pi_1) & \text{add}(0, y) &= y, \ \text{add}(S \ x, y) = S(\text{add}(x, y)). \end{aligned}$$

On total inputs these are genuinely pred, truncated subtraction, min, and add (checked by reduction: $\text{min}(2, 3) = 2$, $\text{min}(3, 1) = 1$, $\text{add}(2, 3) = 5$, etc.). Evaluating their extensions at $x = S^\omega(\perp)$ gives:

$$\widehat{\text{pred}}(x) = S^\omega(\perp), \quad \widehat{\text{sub}}(x, x) = \perp, \quad \boxed{\widehat{\text{min}}(x, x) = \perp} \quad (\text{not } x!), \quad \widehat{\text{add}}(x, x) = S^\omega(\perp).$$

Each equality holds *definitionally* in Agda (`refl`).

The reason is exactly ultimate obstination. Subtraction recurs on its first argument; at $S^\omega(\perp)$ there is no outermost successor to peel, so the Kleene sequence never rises above $\perp$ and $\widehat{\text{sub}}(S^\omega(\perp), x) = \perp$. Since $\text{min} = x \dot{-} (x \dot{-} y)$ is built from subtraction, it inherits this obstination and collapses to $\perp$ on $(S^\omega(\perp), S^\omega(\perp))$. No primitive-recursive algorithm for min can avoid this: it must fully consume an argument, and on $S^\omega(\perp)$ that computation diverges. This is Colson's theorem, made concrete and machine-checked.

Addition is the instructive positive contrast. It, too, recurs on its first argument, but its step function is the *successor*, so the Kleene sequence $u_0 = \perp$, $u_{k+1} = S(u_k)$ is $u_k = S^k(\perp)$, strictly increasing with supremum $S^\omega(\perp)$. This is case 3 with φ strictly increasing, so $\widehat{\text{add}}(S^\omega(\perp), S^\omega(\perp)) = S^\omega(\perp)$: infinity plus infinity is infinity. The denotation faithfully tracks the algorithm — min obstinately diverges to $\perp$, while add climbs to the infinite element.

Files. `Prop1.agda` (Proposition 1), `Property.agda` (the property and `val`), `Computable.agda` (the computability corollary), `PredMin.agda` (the pred/min example), and the recursion chain `USeq*`, `PrecInf*`, `PrecBot*`, `PrecAll`. Source note: `min1.pdf`.